Project Proposal

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Topic: Path Loss Modelling

1. Use the data to build a path loss model, which describes how signal strength decreases as the distance from a cell tower increases.

2. Fit the model to the given data.

3. Investigate whether the same model works for all areas or if different areas require different models.

4. Explain your reasoning and present your analysis in a short report

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1. Problem Statement

The objective of this project is to predict the reduction in signal strength between a Transmitter (*Tx*) and a Receiver (*Rx*) by developing accurate path loss models. The project aims to leverage both supervised and unsupervised machine learning techniques to model and estimate signal attenuation over distance in wireless communication environments.

2. Plan to achieve with the project

Path loss refers to the reduction in the power density of an electromagnetic signal as it propagates through space. Accurate estimation of path loss is critical for efficient wireless network design and optimization.

Objectives:

a. Develop a machine learning-based model to predict path loss using real-world data.
b. Estimate signal strength at various distances from a cell tower.
c. Identify coverage gaps and shadowing regions.
d. Analyze whether a single model generalizes across different environments or multiple models are required.
e. Provide insights useful for optimal cell tower placement and network planning.

3. Project Plan

a. Data Collection: Dataset will be provided by the IIT Madras team. Data is expected to include parameters such as distance, signal strength, location, and environmental factors.
b. Data Preprocessing:
   - Convert raw data into a structured format suitable for machine learning.
   - Handle missing values and remove noise/outliers.
   - Perform exploratory data analysis (EDA).
c. Feature Engineering:
   - Apply feature scaling techniques (normalization/standardization).
   - Encode categorical variables using one-hot encoding.

- Analyze correlation and remove multicollinearity.

d. Supervised/Unsupervised machine learning model development.
e. Model optimization using hyperparameter tuning to improve model accuracy.
f. Model evaluation using methods such as RMSE score, r-square score.
g. Model deployment (*if required*).

4. <u>Tools</u>

a. Python
b. Numpy, scipy, pandas, matplotlib or seaborn (*for plotting*) python libraries
c. Juypter notebook
d. Github (*code repo*)